

Knowledge transfer for global roles in GSE

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Abstract—this practice paper presents how a software engineering organization spread across three countries successfully transferred the knowledge of a few identified roles for a large mission-critical software system that had to conform to regulatory requirements. Multiple releases of the system have been delivered to customers over the 15 years it has been in the market. Each release of the product had a focus area. The competence availability for these focus areas was distributed. As a natural evolution of the globally distributed team, greater responsibility is devolved to a particular location, based on the availability of the competence at that location. Moving the increased responsibility to a location, created a global role, which did not exist earlier. Building the new role required a new skill, what is unique about a global role. Equipping the team members in the new skill was necessary to take up the roles effectively and quickly.

The first step was the identification of the competence for a function/role, training for which may be imparted to another person, who will take over the function/role. This is followed by a process of knowledge transfer, which ensured that a person can take up a new global role from another location. Prioritization based on ease of knowledge transfer for different areas of work, that was found to be effective is described. This helped reduce possible problems that could occur due to incorrect or incomplete transfer of knowledge. The advantages by such knowledge transfer that resulted in new persons taking up global roles have outweighed its disadvantages. The practices described are generic and can be applied to any organization of similar size and complexity.

Keywords—global role, knowledge transfer, product management, test management, test strategy, stake holder involvement, agile, global software engineering, regulatory requirements.

I. INTRODUCTION

This practice paper presents the experience of the team of a globally distributed software engineering organization, which over a period of five years, facilitated competence improvement of a few identified team members in different locations, which in turn helped them in taking over global roles.

This change was considered an essential in order to meet the challenges of achieving significant improvements in development time, getting closer to the customers and having flexible access to specialized workforce.

The competence improvement was done through a structured process of knowledge transfer. The paper describes the identification of the functions/roles that are suitable (or not suitable) for knowledge transfer from one person/location to another and the process of knowledge transfer. The advantages and disadvantages of this process are also described.

II. THE SETUP OF THE GLOBAL TEAM

In our situation, the teams are global and are larger to be only co-located. The distributed teams have a structure of reporting to management at more than one location. The needs of people engagement, supervision, communication and well documented separation of functions/roles within and across teams are met equally irrespective of location.

Each location works on different parts of the overall system and has a team of engineers who have about four to twelve years experience in the domain.

The engineers collaborate to achieve location-specific goals, which are derived from the overall project objectives. The scope of the team is the development and enhancement of a product that must conform to stringent regulatory requirements, with a release duration varying from 3-12 months.

Allocation of work between teams (refer Figure 1) is based on project specific needs and to leverage the advantages of location and the availability of required skills. The teams, composed of 8-10 people scrum teams, with 2-4 week sprint cycle, are expected to accommodate the needs of culture, time zones and use tools to facilitate communication.

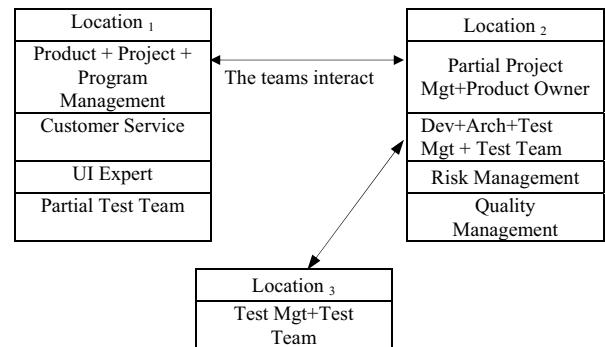


Figure 1: Schematic of team organization

III. THE GLOBAL ROLES

This section describes as to what our global function/roles (Figure 2) are, why they exist and where they exist,

Global role: A global function/role is a role that has engineering responsibility for a particular feature/component of a product globally or a practice area in the global project team.

**Experience paper based on author's experience while he was an employee at Siemens Healthcare Pvt. Ltd, Bangalore, India*

Why they exist: They existed due to a natural evolution of the global teams and are a result of increased co-location of a competence, which over time has taken overall responsibility.

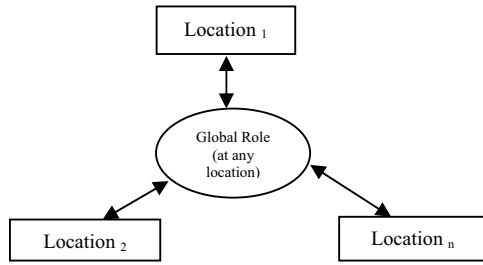


Figure 2 Interface partners of a global role

Collaboration between the roles: The function/roles collaborate with other stake holders [1] within and across the locations/sites to deliver well defined expectations while using communication tools which are common across the locations/sites.

Build up of competence: Equipping the team members in the new skill was necessary to take on the roles effectively and quickly.

IV. MOTIVATION

This section describes the need felt by the organization, while attempting to respond to challenges [2] in GSE. The organization wanted to move more responsibility to global locations.

Global locations are initially set up as outsourcing destinations, from where discrete work packages got delivered.

When these locations grow significantly, in terms of maturity of processes, competence and scalability they attract work, which typically climbs the value chain.

At this time, the senior management teams foresee the possibility of a more meaningful engagement of the location, which may provide solutions to some of the perceived challenges. This facilitates the transfer of knowledge for identified functions/roles and consequently entire ownership of delivery of the project.

The approach to transfer knowledge, which is in general specific to our project, for a role to any suitable destination from parent location, gradually after careful preparation, training, hand holding with an overlap of roles during this period.

Acceptance condition to complete the knowledge transfer process: With an evidence of effective transfer of knowledge and a confidence of continuity of the role at destination location, complete delivery of the function/role from the location would get completed.

V. WHEN THE TIME FOR KNOWLEDGE TRANSFER IS RIGHT

This section provides a general description about the conditions that had an influence on the buildup of the competence.

A. Management directive initiates transfer

The transfer of knowledge for function/role is initiated when there is an overall management guidance to identify, develop a particular function/role at another site for long term engagement.

B. The person with a required competence is hard to find at a location

The time for knowledge transfer is right, when a greater responsibility is moved to a location and there is not enough competence available at that location.

C. The person at a location acquires competence over a period of time

The time for knowledge transfer is right, when a person at partner destination acquires necessary competence over a period of time that matches the expectations of the role within context of a release of the project.

D. Architecture vs organization structure

It was observed by Conway [3] that the structure of a product tends to resemble the structure of the organization that designed it. This typically facilitated more independent development and reduced dependence on interaction between teams.

In our case, it was found that initial design of the product architecture originated at one location. Parts of the dependent components moved to other development centers later. But the roles of the architects remained at the original location, resulting in increasing demands for more collaboration and interaction with remote teams.

This created a suitable condition for the movement of the related role to the remote teams, which predominantly developed and maintained the components.

E. Nature of organizational structure

Evidence from [4] suggests that informal hierarchies rather than a network organizational form offer fewer communication challenges. Quantitative observations [5], have suggested importance of matching organizational structure of the teams involved.

The similarity in the structures of the organizations, competence expectation of the roles and dependency between the stake holders for that role, plays an important part which satisfies a necessary condition for the knowledge transfer for a global role.

This was true in our case. Moreover, as given in [6], switch to matrix based organizational structure of the offshore and onsite teams, due to shared delivery responsibilities, made the collaboration and competence development for the roles possible.

Establishment of collaboration between peers from our partner location also contributed to the creation of a suitable environment for the transfer of knowledge required to perform a role.

Next step in the process is the identification of knowledge components of the function/role. The role and the knowledge

components should be definable, have distinct delivery expectations, interface partners, trainable, and a measurable impact after transfer.

VI. PROCESS FOR KNOWLEDGE TRANSFER

This section provides a process to initiate and complete a knowledge transfer from one location to another. Identified people devote between 50%-70% of time for training purposes.

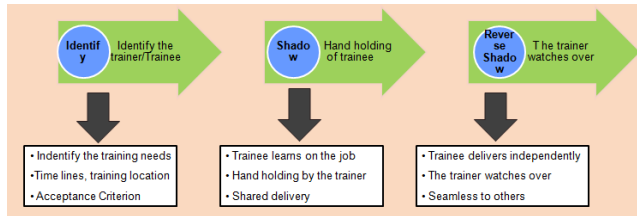


Figure 3 The process steps involved in knowledge transfer

A. Identification of the role/function and its suitability for transfer

The first step involves identifying the function/role that is suitable for knowledge transfer. Before initiating the transfer, the role gets clearly delineated with outcome expectations and well defined interface partners.

B. Knowledge transfer

The persons who facilitate the knowledge transfer at source and destination sites are identified. The knowledge transfer is initiated by establishing communication between the two.

It is possible that the persons involved visit each other's location for a period that is considered necessary to facilitate an effective interpersonal communication for the transfer of knowledge.

Any competence development task that does not require face-to-face communication and can be trained remotely was done remotely by using web based communication tools.

The task of knowledge transfer is considered as one of the overall project goals for a product release. The actual transfer of knowledge once initiated, follows the regular project management practices of planning, tracking and reporting.

C. Overlap of roles – important step

Once the person acquires a reasonable competence in the expected role, there is a period of overlap called 'shadowing', where both the persons play the role interchangeably. This provides required 'on-the-job' training, which would facilitate handling the function/role independently later.

D. Evaluation of independent functioning of the role

Once the knowledge transfer process is completed, an evaluation is done, if the receiver can function independently. A phase, called 'reverse shadowing', where the giver of the knowledge transfer, watches over the delivery of the role by the receiver of knowledge transfer.

E. Complete delivery from destination – last step of the process

Once it is agreed mutually that receiver person can function independently, the role is delivered fully without any support from the giver of the knowledge transfer.

The above process would complete the process steps for the movement of the roles.

VII. OPTIMAL SEQUENCE OF KNOWLEDGE TRANSFER AND REASONS FOR IT

This section describes the prioritization or sequence of knowledge transfer for practice areas, with shared roles, which was found most effective in our context. Refer Table below.

Practice Area	Pre-requisite + Mode of training	Acceptance condition
A.Configuration Management	Common s/w processes and tools + integrated dev. tools +group awareness and adherence to the rules on the management of configuration elements. <i>Training is provided in person on the use of common tools +visit to different sites + use of common mail groups</i>	Entire configuration management is from one location and seamless to users
B.Continuous Integration	Existence of relevant hardware and s/w tools + coupling to the configuration management and s/w dev tools <i>One person trained in person and on site + Offline training to configuration management team + dev +architects</i>	Entire continuous integration is from one location and is seamless to the users

Table 1 The table of practice area and the mode of training

A. Configuration management function

Early introduction of common software processes across different sites, global group awareness of software configuration elements, verbal agreements at an early stage of development, integrating software development tools with configuration management environment and the use of similar hardware made it easy to move configuration management function across sites.

The fact that the tools used for configuration management themselves support multi-site development facilitated the seamless transfer of ownership and control from any site.

B. Continuous Integration

The reasons given above for configuration management are applicable for this practice area as well.

The table below describes the pre-requisite, training provided and acceptance criterion for other functions like project management, product management, test management and software development functions.

Function Area	Pre-requisite + Mode of training	Acceptance condition
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C. Increased development responsibility	The components with higher development content for a release is owned by a particular site + availability of large pool of s/w developers at the site + capacity to scale up+ strict code quality monitoring mechanism <i>Class room and remote/offline training for identified dev teams + use of common coding guidelines + roll out of process and use of tools that ensure code quality</i>	Higher proportion of usable s/w components produced with high quality and within agreed time frame
D.Product Owner/System Analyst (PO/SAN)	Increased competence on the domain/application at a site + need to station a PO/SAN at a site, based on geographic nearness considerations <i>Trained by product managers+ visit to hospitals +interactions with service personnel + application specialists + technical documentation skills + intended use +involvement in daily stand up</i>	Product features accepted by end users without major complaints
C.System Architect	Excellent knowledge on product design and architecture + long time architect +organization certified architect + decision making <i>Trained by senior architects at parent site + hand holding, involvement in all architecture decision + use of common s/w dev tools +product usage</i>	There are no major architecture issues found during development or after a release of a product
E,F.Project management (Equally applicable for Test management)	There is a need felt for such a shared global role +the person identified has adequate experience in project management + industry certified /PM@organization certified <i>Hand holding by a team of senior project managers at different sites + short term stay at parent site with shared delivery responsibility +process orientation+ awareness training on other functions</i>	The satisfactory shared delivery for at least one product release + motivate the team towards common goals + gets other stake holders support

Table 2 the knowledge transfer required for a global role

C. Increased development and system architect responsibility

Third function that was considered for transferring the knowledge was for a larger amount of development responsibility. This was because of abundant availability of the qualified and competent development team member at the site. Also a structured training facility that moulds the team member in the required competence also contributed to it. Over a short period of time, as a natural consequence of increased development responsibility, the system architect role was also transferred with increased competence and experience in the same product area.

D. Partial product management function

Nearness to the customer location, made it difficult to transfer the knowledge required for the complete product management function. However knowledge and training required for some supporting roles was provided, once it was

noticed that there is an increased competence at the site. Due to this the employees from more than site, could support the customer needs globally.

E. Complete test management function

In agile development methodology followed in our project, test team members interact very closely with development team members to provide immediate feedback on the components developed.

When higher proportion of development work is from one location, there is a possibility of an improved productivity and faster feedback and delivery if a higher proportion of testing work also gets done from the same location. Though this may not be true always, but in our context it proved more effective. This also facilitated a more unified and integrated test strategy, when changes to product code and the subsequent test approach were discussed and refined on a daily basis.

F. Partial project management function

Being part of the same organization, similar organizational structure contributed to creation of roles that performed functions which could be called peer roles across the different sites. Some sites had project leads, who supported the overall project manager. Broadly project management developed as a group of people, who were supported by their peers and guided on strategic and daily operational purposes.

As a software product under development, which underwent regular release cycles, the type of issues faced per release were similar. Awareness of the likely issues along with intense collaboration between peers, during their resolution, provided an opportunity for the project management teams at different location to acquire adequate competence. This created a favorable condition for the responsibility of project management to get shared across sites.

However only a partial sharing of the function was possible, as the parent location only gave a unified strategic direction and permitted the knowledge transfer.

VIII. COMPLETE KNOWLEDGE TRANSFER FOR THESE ROLES MAY BE INCORRECT

This section describes the functions that have been found to give optimal results when not transferred completely.

A. Partial product management function

The product management has to work closely with market needs. If a region has several installation of a product, it is likely that there would be a need for a faster response to the requirements of the customers/users there. This is best done within the time zone and in the native language, e.g like in Europe. Due to this reason, some people may be trained to perform product management functions at more than one site, but it is not generally recommended to group all of them at one location only.

B. Partial project management function

Our product being a legacy product which has a large installed base, there is a need for project management to be located centrally, to give strategic orientation and for long term product goals/vision. The project management has to work

very closely with product management to facilitate the realization of these goals. This collaboration is more likely to succeed, in case both these functions are co-located like in our situation, than when they are distributed.

C. Language testing function

For product like ours, that is available in multiple languages, expectation of a competence to test it in all those languages at any one site may not be justified. For example, as our product is released in many European languages, it is essential to retain language-trained validation engineers at different sites, within Europe.

D. Customer service function

As our product is available in several regions of the world, it became necessary to deploy customer service personnel closest the region of installation. This facilitated a faster service, which would not be possible, if these personnel were to travel from distant remote location. This also has a monetary implication, when the customer service personnel travel long distances, costing large amounts of money, for servicing the products.

IX. TOOLS USED, ADVANTAGES AND DISADVANTAGES OF A KNOWLEDGE TRANSFER

The tools used are in commercially available communicator tools. Novelty of the approach is the fit people for roles rather than roles fitting people.

A. Advantages/benefits

1) Talent/skill available is leveraged

It was possible to achieve faster and better resolution of technical and management issues. When the group of product owners is co-located with the development team, they could evaluate and provide feedback on the product features while they are getting developed. Participation of test management peer group in the daily stand up meetings helped align the testing approach.

2) Reduced dependence on a single site

An effective back up was created, when technical skill of the people at the destination location took up a global role. It is no longer necessary to depend on availability or time zone difference for a decision. Occasional work overload on a particular person is reduced. Due to availability of multiple people with adequate competence for a role at more than one location, any overwhelming dependence on any one site or any one individual may reduce. This situation is more desired as it can provide continuity and broader spread of knowledge, essential for a product like ours.

3) Career path at destination site

It provided a career path for employees who took up the roles, which increased the motivation and better outcomes.

4) Cost arbitrage

Though cost reduction was not the only goal, the transfer of knowledge within and across locations, has reduced the cost (measured in terms of time) of acquisition of a new employee

by almost 50%, as compared earlier and also due to the fact that existing people took up the roles, at no extra cost.

B. Disadvantages

1) Incorrect role assignment/incomplete transfer

In such occasions, where the identification of person to receive the knowledge is incorrect or the knowledge transfer itself is ineffective, there could be inadequacy with consequent delivery of the role expectations. For example, the expectation of a test architect role to mentor manual and automation teams simultaneously. This may not be fulfilled, as it requires different types of competencies, and it may not be performed effectively at the same time, without adequate up-skilling of the test architect.

2) Remote chance of loss continuity or direction

There is a chance of loss of continuity or direction of project goals, in case of differing perspectives of people, if the person receiving the knowledge differs while collaborating with the person currently playing the same role. This can create a need for additional training and orientation.

3) People challenges at a remote site

The people manager when located at a remote site can potentially delay or ineffectively resolve people issues. Resolution of people issues is less effective if the people manager is located at another site. The use of different types of communication tools is not a substitute for face-to-face interaction. There are instances, where a manager had to travel to another location to resolve such issues.

X. SUMMARY AND CONCLUSION

The identification of competence for function/role, suitable to be trained or not coupled with structured process of knowledge transfer, helped achieve a goal of leveraging the talent at different locations, and made the product delivery more sustained and predictable, as compared to ad-hoc approach adapted earlier.

These practices are generic not only for our application domain alone and can be applied to any project of similar complexity, by adapting them to its processes. The rigor in these practices helped manage the expectations of the key stakeholders.

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